HAIP V1 — Polish Pass

Optimize the existing product. No new modules.

Product / Engineering

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Optimize the existing product. No new modules. Target: 90 days. ~80 fixes ranked by ROI. **Date:** May 2026

How to use this memo

This is the opposite of the V2 PRD. There are no new modules here. Every item is a fix, polish, or hardening of something that already exists in V1. Each item has:

- **Where** — actual file path or page name from the V1 codebase
- **Friction** — what hurts today
- **Fix** — what to change
- **Effort** — S (hours-1 day), M (2-4 days), L (week+)
- **Impact** — 1-5 (5 = users feel it every day, 1 = nice-to-have)

The top 15 are at the front. The rest are organized by module. The last section is a 30/60/90-day sprint plan and metrics to watch.

A note on philosophy: most of these fixes are not glamorous. None of them will get a tweet. All of them are felt by a real operator on a real Tuesday at 7pm when they are tired and trying to close a booking. The compounding effect of polish is the difference between a tool people tolerate and a tool people love. That's what this memo is for.

Part I — The Top 15

These are the items that will move the needle most in the next 30 days. Picked on impact × effort × user-felt-daily.

#	Fix	Module	Effort	Impact
1	Idempotency keys on every outbound send (WA, IG, SMS)	Backend hygiene	S	5
2	Surface message delivery status (sent → delivered → read) with visible color states	Inbox composer	S	5
3	Stage transition validation (no NEW → BOOKED jumps)	Inquiry CRM	S	5
4	Webhook signature verification — require Meta app secret in prod (fail-closed)	Security/ backend	S	5
5	Saved replies / canned snippets with / trigger + {{guest_name}} variables	Inbox composer	M	5
6	SLA stopwatch + breach badge on Kanban cards and inbox conversations	Inquiry CRM	M	5
7	Click-any-word in transcript → seek audio + persisted speed	Call detail	M	5
8	Inbox sorting by unread + SLA-urgent	Inbox list	S	4

	first, not just last-message-time			
9	Per-tenant timezone + business hours (kill hardcoded IST)	Backend / settings	M	4
10	Auto-detect opt-out keywords (STOP / UNSUBSCRIBE / बंद करो) + propagate suppression	Compliance	M	5
11	Calls list pagination + server-side filters (handle 1000+ rows)	Calls list	M	4
12	Celery task retry strategy (Meta, Deepgram, transcription) with dead-letter visibility	Backend hygiene	M	5
13	Cmd+Enter to send everywhere; j/k to navigate inbox; ? shortcut help overlay	Cross-cutting UX	S	4
14	Sample-data toggle on first login (20 fake convos, 10 calls) for empty tenants	Onboarding	M	4
15	Undo toast (8 seconds) for archive / delete	Cross-cutting UX	S	4

		/ lost-stage actions			

If you ship only these 15 in 30 days, your weekly tenant NPS will move. The rest of this memo is what comes after.

Part II — Polish Principles (the four)

Before the per-module lists, four principles that should govern every decision:

1. Visible status beats fast operations. A green dot saying "WhatsApp connected, last synced 12s ago" is worth more than a 50ms latency improvement. Operators tolerate slowness if they trust the state. They never tolerate ambiguity.

2. Every destructive action needs an undo. Archive, delete, mark-lost, stage-move — all of these reach a moment where a tired operator clicks the wrong thing. 8-second toast undo is the cheapest insurance HAIP can buy.

3. Empty is the most expensive state. The empty inbox on day 1 of a new tenant is where most of your churn lives. Every empty state needs an action and ideally sample data.

4. Compounding small wins beat one big shipment. Ten S-level fixes shipped in a week is worth more than one M-level feature shipped in a month — because each S-fix is observed daily by your power users, and they each create a "the product got better this week" signal.

These four principles are why the rest of this list exists.

Part III — Module-by-module polish

1. WhatsApp Inbox & Conversations

The inbox is where your power users live. A 30% improvement here is felt 100x more than a 50% improvement anywhere else. These are ranked by impact.

Composer

#	Where	Friction	Fix	Effort	Impact
W1	inbox/page.tsx:282	Enter sends even on empty input	Disable send unless value.trim().length > 0; show pulse on send	S	4

W2	inbox/page.tsx:285	Shift+Enter behaves oddly on mobile, no multi-line	Fix textarea key handler; auto-grow on newlines; show “Shift+Enter for newline” hint	S	4
W3	inbox/page.tsx composer	No / trigger for saved replies	Add saved-reply menu on / with fuzzy search; merge {{guest.name}}, {{property}}, {{check_in}}	M	5
W4	inbox/page.tsx composer	No scheduled send	Add clock-icon → “send tomorrow 9am” / custom datetime picker	M	4
W5	inbox/page.tsx composer	No reply-to-specific-message (quote)	Right-click message → “Reply”; outbound carries quoted reference	M	4
W6	inbox/page.tsx composer	No paste-image-from-clipboard	Listen for paste event, upload + preview	S	3
W7	inbox/page.tsx composer	AI draft button confidence shown but no “regenerate” / “tone” controls	Add tone chips (warm / formal / brief) + regenerate button; preserve last	M	4

			draft on reject		
W8	inbox/page.tsx:228-246	Rejected AI draft is gone forever	Stash last 3 rejected drafts; small "↶" to recall	S	3
W9	Backend messaging.py	No idempotency on send — retries can duplicate	Add Idempotency-Key header support; backend dedups on key+tenant for 24h	S	5
W10	inbox/page.tsx composer	Markdown shortcuts not previewed	Show *bold* → **bold** preview live; WhatsApp-native syntax	S	3

Conversation list

#	Where	Friction	Fix	Effort	Impact
W11	components/inbox/ConversationList.tsx:95	Unread badge caps at 9+ but list doesn't surface unread to top	Sort: unread → SLA-urgent → last-message; preserve manual pin	S	5
W12	ConversationList.tsx:70	Truncated last-message preview, no tooltip	Tooltip on hover; truncate to 80 chars	S	3
W13	ConversationList.tsx	No multi-select for bulk archive/assign/tag	Click + shift-click to range-select; bulk action bar appears	M	4
W14	ConversationList.tsx	No snooze	Right-click → snooze (3h / tomorrow / next week /	M	5

			custom); show snoozed badge		
W15	ConversationList.tsx	No saved views (“Unassigned mine”, “Awaiting reply >2h”)	Add view-saver; default views ship with product	M	4
W16	ConversationList.tsx	No SLA timer per conversation	Show “1m left” → “BREACH +12m” badge	M	5
W17	ConversationList.tsx	No conversation search (full-text)	/ to search; filter by date / agent / tag / status	M	4
W18	ConversationList.tsx	No “agent X is replying…” presence indicator	Socket.io broadcast typing state; show collision warning on send conflict	M	4
W19	ConversationList.tsx	No tags / labels with color	Add Tag entity (already partial); color chips in list + composer-side tag editor	M	3

Conversation detail thread

#	Where	Friction	Fix	Effort	Impact
W20	MessageBubble.tsx:23-30	Message status icons silent until hover	Always-visible single → double → blue check; failed = red retry icon	S	5

W21	inbox/page.tsx:301-335	Right-rail guest profile, no “open in new tab”	Add icon; meta-click opens in new tab	S	3
W22	Composer area	No drag-drop file upload visible affordance	Add dashed border on dragover + drop zone	S	3
W23	Composer area	Sending failure shows a toast that vanishes; can’t retry from message	Inline retry button on failed messages; visible 24h	S	4
W24	Conversation thread	No internal notes / private comments	Add /note slash command; notes render with yellow background, never sent to WA	M	4
W25	Conversation thread	No @mention of teammate	@ opens user picker; mention pushes notification	M	4
W26	Conversation thread	No emoji-reaction support on inbound messages	Add reaction picker (WA supports natively)	S	2
W27	Conversation thread	No voice-note playback speed	Add 1x/1.5x/2x toggle on inbound voice notes	S	3

2. Inquiry CRM (Kanban + Pipeline)

The CRM is where your sales team lives. Stage hygiene + visible ownership = the whole game.

#	Where	Friction	Fix	Effort	Impact
K1	services/inquiry_service.py:39-53	Any stage transition allowed (NEW → BOOKED)	State-machine validation: allowed transitions + reason on backward moves; audit log entry	S	5
K2	inquiries/page.tsx:113-137	No drag-drop visual feedback	Lift card on grab; highlight valid drop columns; flash on invalid drop	M	4
K3	inquiries/page.tsx:28-34	STAGES hardcoded; can't hide / reorder	Per-tenant stage config; hide unused; reorder columns	M	3
K4	components/crm/KanbanCard.tsx	No assignee avatar on card	Show assignee initials chip; unassigned = ⚠ red chip	S	5
K5	KanbanCard.tsx	No lead-age signal — stale leads buried	Color-code card border: <24h green / 1-3d yellow / >3d red	S	4
K6	KanbanCard.tsx	No SLA breach indicator	Show SLA stopwatch chip with breach color	M	5
K7	inquiries/page.tsx:14	Table view doesn't	Add columns to table	S	4

	0-173	show unread or SLA	view; persist column choices		
K8	inquiries/ page.tsx:78 -80	Cards within column not sorted (stale at top)	Sort: SLA-breach → stale → recent within each column	S	4
K9	inquiries/ page.tsx:55 -75	Lost-reason dropdown forces hardcoded reasons; no free text	Allow “Other” with required free-text; cluster these for analytics	S	4
K10	inquiries/ page.tsx	No bulk actions (assign 20 leads to rep)	Multi-select on table view; bulk reassign / tag / move	M	4
K11	inquiries/ page.tsx	No saved filters per user	“My open leads in Goa” save view; per-user persistence	M	3
K12	inquiries/ page.tsx:86	“Drag cards to update stage” hint only shows once	Persist hint in localStorage; show once per user, dismissible	S	2
K13	inquiries/ page.tsx	No inquiry-aging report inside CRM	Small widget on Kanban header: “8 leads stale >3 days”	S	4
K14	Backend inquiry_ser vice.py	No assign_roun d_robin action	Simple round-robin within team; respects out-of-office toggle	M	4
K15	Inquiry	No internal	Add	M	4

	detail	notes thread per inquiry	InquiryNote entity (already exists in audit?) + UI with @mentions		

3. Call Intelligence (Detail + List)

This is V1's strongest module. The fixes here are sharpening, not building.

Call detail page

#	Where	Friction	Fix	Effort	Impact
C1	calls/[id]/page.tsx:355-374	Click on transcript turn — audio seek drifts	Fix currentTime math (use turn start, not turn index); add 200ms snap tolerance	S	5
C2	components/calls/CallPlayer.tsx:79-167	Fake sine-wave bars, not real audio waveform	Use wavesurfer.js (~30KB); render real peaks at S3 fetch	M	4
C3	calls/[id]/page.tsx:394-488	QA evidence line-clamp-1 hides text	Expand on click; show full evidence inline; copy-to-clipboard button	S	4
C4	calls/[id]/page.tsx:214-262	No manual refresh on stuck transcription	Add "Refresh status" button; show last-poll timestamp	S	3
C5	calls/[id]/page.tsx:99-100	"Show all criteria" toggles all-	Per-criterion expand/collapse	S	3

		or-nothing			
C6	calls/[id]/page.tsx:369	Click on turn doesn't pause if audio elsewhere	Set currentTime + ensure play state; toggle on second click	S	4
C7	Call detail	No inline coaching comments	Allow manager to drop a note at any timestamp; agent sees them in their feed	M	4
C8	Call detail	No bookmark / star moments	One-click bookmark on transcript turn; bookmarks list in side panel	S	3
C9	Call detail	No "share clip" link (range, view-only)	Generate signed URL with start/end; renders mini player on landing	L	3
C10	Call detail	No playback speed (or speed not persisted)	Add 1x/1.25x/1.5x/2x; localStorage persist per user	S	4
C11	Call detail	No keyboard shortcuts (space play/pause, ←→ skip 5s)	Implement on focus; show in ? overlay	S	4
C12	Call detail	No transcript search-in-call	/ search; result count + jump-to	S	3

C13	Call detail	No agent talk-ratio bar	Single bar: agent% / guest%; computed from diarization	S	4
C14	Call detail	Missing-revenue badge has no drill-down	Click → show evidence + suggested coaching action	M	3
C15	Call detail	Sentiment shown as single label, not trajectory	Mini line chart of sentiment over time; jump-on-click	M	3

Calls list

#	Where	Friction	Fix	Effort	Impact
C16	Backend calls.py list endpoint	No pagination — returns all rows	Server-side pagination (cursor preferred); 50 per page default	M	5
C17	calls/ page.tsx:73 -92	Filters apply one at a time; no saved views	Apply multiple at once with “Apply” button; save view	S	4
C18	calls/ page.tsx:94 -100	Search is client-side, case-sensitive on phone	Server search with E.164 normalizatio n; “987654321 0” matches “+91987654 3210”	S	4
C19	calls/ page.tsx:17 8-192	Wrap-up queue not separately	Pin “Needs wrap-up” smart view	S	4

		visible	to sidebar with count badge		
C20	calls/ page.tsx:209-265	No QA score trend column (▲/▼)	Compute from rolling 5; show arrow	M	3
C21	calls/ page.tsx	No bulk actions (tag, mark reviewed)	Multi-select + bulk action bar	M	3
C22	Calls list	No “review needed” filter (e.g., score < 60 + not reviewed)	Smart view with count badge	S	4

4. Guest Detail & Timeline

#	Where	Friction	Fix	Effort	Impact
G1	guests/ [id]/ page.tsx:20-32	Timeline shows all events, no filter	Toggle chips: Messages / Calls / Stays / Inquiries / Notes	S	4
G2	guests/ [id]/ page.tsx	No revenue summary (total spend, last spend, LTV)	Header card: total spend / # stays / last visit / repeat-flag	M	4
G3	guests/ [id]/ page.tsx	No “days since last contact”	Header chip “Last contact: 47 days ago”	S	3
G4	guests/ page.tsx:40-49	Search debounce 300ms feels stale while typing	Reduce to 150ms + cancel in-flight requests	S	3
G5	Guest detail	No quick-	Header	S	4

		action buttons (new inquiry, call now, send WA)	action row		
G6	Guest detail	Past stays don't show OTA vs direct source	Add source badge + commission impact	S	3
G7	Guest detail	No "merge guests" UI	When two phones suspected same person, suggest merge; admin approves	M	4
G8	guests/ [id]/ page.tsx:49 -58	DPDP delete is one-click with confirm; no soft-delete buffer	30-day grace period; show in admin queue with restore option	M	4
G9	guests/ page.tsx	No filter on tags, has-stay, has-call, last-active	Add filter pills	S	3
G10	Guest list	No "days since last contact" column	Add sortable column	S	3

5. Templates / Campaigns / Subscribers / Flows

These pages were promised in V1 but largely empty. Polish here is honest UI — either ship a real minimum-viable surface, or be clear about coming-soon.

#	Where	Friction	Fix	Effort	Impact
T1	templates/ page.tsx:98 -115	Create-template	Add draft mode;	S	4

		modal — no “save draft”	persist between sessions		
T2	templates/page.tsx	Rejected templates have no “resubmit” — must recreate	Add “Edit & resubmit”; preserve history	S	4
T3	templates/page.tsx:43-51	Language dropdown limited; no warning on language mismatch with content	Add language detection on body; warn if mismatch	S	3
T4	templates/page.tsx	No pre-submit linter for likely rejection reasons	Lint: URL shorteners, promotional language in UTILITY, too many variables, missing samples	M	5
T5	templates/page.tsx	Live preview doesn’t match real WA rendering	Use exact WA-iOS / Android styles; test header/body/footer/buttons	M	3
T6	campaigns/page.tsx	Currently a “coming soon” stub	Ship a minimum send-broadcast-to-CSV-list feature first (audience builder is V2 scope)	L	4
T7	subscribers/page.tsx	Likely no CSV bulk-import	Add CSV upload + dedup	M	4

			against existing guests		
T8	flows/ page.tsx	Coming-soon stub	Add a single hardcoded “Pre-arrival 48h” Flow that runs on every confirmed Stay; defer Studio to V2	M	5

6. Analytics & Reports

#	Where	Friction	Fix	Effort	Impact
A1	analytics/ page.tsx:15 -32	KPI cards have no drill-down (“p95 = 300s” — bad? good? where?)	Click → side-panel with breakdown by Property, by agent, by hour	M	4
A2	analytics/ page.tsx	No comparison view (this week vs last)	Toggle “compare to previous period”; render delta %	M	4
A3	analytics/ page.tsx	No per-Property breakdown	Add Property filter at top; persists across charts	S	4
A4	analytics/ page.tsx	Lost-reason themes pie chart, no drill-into-theme	Click slice → list of inquiries with that reason	M	4
A5	analytics/ page.tsx	No export to CSV	Add per-chart export button	S	3

A6	Backend analytics	Lost-reason clustering runs full table weekly	Incremental: cluster only inquiries closed since last run	M	3
A7	Analytics	No “daily owner digest” via WA/email	Cron at 9am: top 5 metrics + 1 insight; opt-in	M	4
A8	Analytics	No “anomaly highlight” on charts (subtle dot for outlier days)	Mark days >2σ from rolling mean; tooltip on hover	M	3

7. Onboarding

V1 onboarding is 9 steps. Hospitality SMB operators give you 5 minutes before bouncing.

#	Where	Friction	Fix	Effort	Impact
O1	onboarding / page.tsx:19	9 required steps	Make steps 4-9 skippable with “Set up later” → unlock from settings	M	5
O2	Onboarding	No sample-data toggle	After step 3, offer “Try with sample data” → 20 fake convos + 10 fake calls; one-click wipe	M	5
O3	onboarding / page.tsx:37-48	IVR/PMS choices unranked, no help text	Sort by Indian market share; add “Not sure?” link → contact us	S	3

			via WA		
O4	Onboarding	No upfront validation; nexting throws server errors	Validate per-step; show errors inline before Next	S	4
O5	Onboarding	No in-product checklist post-onboarding	Persistent bottom-right widget: Connect WA / Invite teammate / Send first reply / View first call — confetti on complete	M	5
O6	Onboarding	No “test connection” after integrations	Each integration: green check + last-synced timestamp + retest button	S	4
O7	Onboarding	No phone+OTP signup option	Add WhatsApp-OTP login for Indian SMB	M	3
O8	Onboarding	Default DPDP consents bundled, not granular	Separate checkboxes per purpose (transactional / marketing / analytics)	S	4
O9	Onboarding	Default empty inbox after onboarding	Pre-populate 1 sample conversation with explainer (“This is what a guest	M	3

			message looks like"); auto-archive on first real one		
O10	Welcome flow	No "Invite teammate via WhatsApp"	One-click WA share with team-join link	M	4

8. Settings & Integrations

#	Where	Friction	Fix	Effort	Impact
S1	settings/ integrations/ page.tsx:18-46	Error state hidden behind card expand	Always-visible error message + "Reconnect" CTA	S	4
S2	settings/ integrations	No "test connection" after re-entering creds	Add test-now button → returns success/failure synchronously	S	4
S3	settings/ integrations	No "last synced" timestamps	Display per-integration last-sync time and status	S	4
S4	settings/ team	No per-property role assignment	Add property scopes per role; agent on Villa A can't see Villa B	L	4
S5	settings/ branding	No live preview of brand colors	Theme tokens applied live in preview pane	S	3
S6	settings/ billing	No self-serve	Phase 1: link	M	3

		plan change	to email support with prefilled subject; Phase 2: real billing UI		
S7	settings/ integrations	Meta WABA quality rating not shown	Pull from Meta API; show green/yellow/red with link to fixes	M	4
S8	settings	No keyboard shortcut to find a setting (search)	Cmd+K → fuzzy search settings	M	3
S9	settings/ integrations	No webhook delivery log	Show last 50 webhook deliveries + failure reasons	M	4

9. Compliance, DPDP & Meta Policy

These are not big features. They are insurance.

#	Where	Friction	Fix	Effort	Impact
D1	Compliance/ messaging	Opt-out keywords not auto-detected (STOP / UNSUBSCRIBE / बंद करो / रोको)	Add inbound message keyword detector; auto-flag Consent.marketing=false; send confirmation	M	5
D2	Compliance	Opt-out doesn't propagate to broadcast lists	On opt-out, suppress in next broadcast within 30s; propagate WA → IG → email	S	5

D3	Compliance	No consent proof storage (timestamp, IP, UA, version of notice)	Persist in Consent table on every capture	M	5
D4	Compliance	Notice not bilingual (DPDP requires regional language)	Add Hindi (and Marathi for Maharashtra tenants) versions of notice	S	4
D5	Compliance	Retention auto-purge not implemented	Nightly job: delete transcripts >24mo, calls >12mo, messages >36mo (per tenant config)	M	4
D6	Compliance	Grievance officer not in footer	Add static footer + dedicated /grievance form with response SLA tracker	S	4
D7	compliance/page.tsx:50-98	Audit log shows action name, not resource	Add resource_type + resource_id; clickable to the row	S	4
D8	compliance/page.tsx:64-68	Consent registry silent on fetch failure	Replace .catch(() => []) with error toast + retry button	S	4
D9	Compliance	No 24h-window indicator on	Show timer: “Service window:	M	5

		conversations	18h 22m left"; block free-form send + auto-suggest template after 24h		
D10	Compliance	Template linter doesn't flag URL shorteners or category mismatch	Pre-submit lint with reason codes	M	4
D11	Compliance	DPDP delete: one-click with no grace period	30-day soft-delete buffer; admin queue with restore	M	4
D12	Compliance	No breach notification template	Pre-draft 72-hour notification template + incident form for admin	S	3
D13	Compliance	No PII redaction in logs	Mask phone, email, Aadhaar in app logs (Presidio + custom patterns)	M	4

10. Mobile & Responsiveness

V1 is desktop-first. Mobile-friendly = retention from owners checking their phone at dinner.

#	Where	Friction	Fix	Effort	Impact
M1	inbox/page.tsx:40-45	Channel filter horizontal scroll on	Add scroll-shadow fade on right edge; or	S	3

		mobile, no overflow indicator	stack vertically <640px		
M2	calls/[id]/page.tsx:313-392	Score ring takes full width on mobile, overwhelming	Collapse score panel to expandable card below transcript on mobile	S	3
M3	inquiries/page.tsx:103-137	Kanban horizontal scroll on mobile	Add scroll-snap per column; show column indicator dots	M	3
M4	Cross-product	No bottom-tab navigation on mobile	Add 4-tab nav (Inbox / Inquiries / Calls / More) under 640px	M	4
M5	Cross-product	Tap targets too small (icons <40px)	Audit all icon buttons → min 40×40 on mobile	S	3
M6	Cross-product	Modals too tall, no scroll-trap	Audit modals; ensure body scroll-lock + max-h-90vh	S	3
M7	Inbox	Keyboard covers composer on iOS	Add viewport-fit=cover + safe-area-inset for iOS bottom bar	S	3

11. Backend Hygiene & Tech Debt

The audit found ~10 quiet liabilities. Fix them before they bite at scale.

#	Where	Friction	Fix	Effort	Impact
B1	core/config.py:78 + webhook handlers	Webhook signature skipped if Meta secret unset	Fail-closed in prod if META_APP_SECRET missing; deploy check	S	5
B2	inquiry_service.py:39-53	No stage transition validation	State machine; backward moves require reason + audit entry	S	5
B3	All Celery tasks (workers/tasks/*)	No structured retry with backoff	Add autoretry_for + exponential backoff; dead-letter queue with visibility	M	5
B4	messaging.py send path	No idempotency keys; retries can duplicate WA messages	Idempotency-Key header → backend dedup 24h	S	5
B5	conversations.py:49-81	Worker delivery failure silently leaves was_approved=True	Track delivery state separately (queued / sent / delivered / failed); update on worker callback	M	5
B6	core/config.py:18	ALLOWED_ORIGINS defaults to localhost;	Refuse to start in production env if	S	4

		permissive in prod if not changed	ALLOWED_ORIGINS not explicitly set		
B7	core/database.py:27	RLS tenant_id set via string interpolation (SET app.tenant_id = '{tenant_id}')	Use parameterized SET LOCAL via text() with bound param	M	5
B8	API endpoints	No rate limiting (e.g., /calls, /conversations list)	Add slowapi / fastapi-limiter middleware; per-tenant + per-IP	M	4
B9	message table	No partitioning strategy; unbounded growth	Partition by month + tenant_id from new tenants forward; backfill plan	L	3
B10	Cron schedules	Hardcoded IST	Per-tenant timezone + per-tenant business hours config; use APScheduler with tenant context	M	4
B11	Lost-reason clustering	Full-table scan weekly	Incremental: cluster only newly-closed inquiries since last run; merge centroids	M	3
B12	Test	No mocks	Add respx	M	4

	coverage	for Meta / Deepgram / Claude	or pytest-httpx fixtures; record-replay for transcription		
B13	Test coverage	No E2E test for WA webhook → Inquiry → Conversation → Message	Add pytest integration test that walks the chain end-to-end	M	5
B14	Test coverage	No Celery chain retry test	Test transcription failure → retry → fallback adapter	M	4
B15	Recording fetch	No fallback if S3 object missing	On 404 → mark Call.recording_status='missing' + retry job after 1h; do not crash transcription chain	S	3
B16	All endpoints	No request_id propagation for trace	Add X-Request-ID middleware; surface in error responses; log everywhere	S	3
B17	Sentry	No release tagging	Tag releases in Sentry on deploy; auto-link errors to git SHA	S	3
B18	Postgres	No slow-query log	Enable pg_stat_stat	S	3

		review process	ements; weekly top-10 review		
B19	Redis	No memory eviction policy explicit	Set maxmemory-policy allkeys-lru; alarm on >80%	S	2

12. Cross-cutting UX polish

These apply everywhere; ship as a shared component update.

#	Friction	Fix	Effort	Impact
X1	Destructive actions have no undo	8-second undo toast for archive / delete / lost / bulk actions	S	5
X2	No keyboard shortcuts	j/k navigate lists; e archive; r reply; Cmd+Enter send; / search; ? overlay	S	4
X3	Spinners everywhere	Replace with shadcn skeleton components	S	3
X4	No optimistic UI on sends / status changes	Render new message immediately, gray until ack; revert on failure	S	4
X5	Inconsistent date display	Single <RelativeTime> component: “2m ago”, “yesterday 4pm”, “May 8”	S	3
X6	No command palette	Cmd+K: jump to guest /	M	4

		conversation / page / setting (fuzzy)		
X7	No consistent empty states	Audit every list view; ensure CTA + 60s Loom embed; no blank screens	M	4
X8	No global status bar	Top-of-app pill: “WhatsApp: green / IVR: green / AI: 12s p50”; click → detail	M	4
X9	No “changelog in-app”	What's new icon in header; modal with last 5 release notes	S	3
X10	Help button — currently?	In-app Loom library 60-90s per topic + WA support button (highest-trust signal in India)	M	4
X11	No tooltip discovery	Hotspot pulse on first session for top 5 features; dismissible	S	3
X12	Form labels overloaded	Standardize: label / help text / error pattern; one component	S	2
X13	No Esc to close modal everywhere	Audit modals; ensure consistent	S	3
X14	No focus rings (accessibility)	Restore default focus rings; ensure keyboard nav visible	S	3
X15	Inconsistent	Audit	S	2

button hierarchy	primary/secondary/destructive button usage

Part IV — 30 / 60 / 90 Day Sprint Plan

Take this as a sequencing recommendation, not a rule. Reorder based on customer feedback.

Days 1–30 — Trust + Speed

Theme: *fix the silent failures, give users keyboard speed.* High-impact, low-effort, every-day-felt.

- Top 15 (Part I) — all of them.
- Plus from cross-cutting: X1 (undo), X2 (keyboard), X3 (skeletons), X4 (optimistic UI), X5 (dates).
- Plus backend hygiene: B1 (webhook sig), B2 (stage validation), B6 (CORS), B7 (RLS injection fix), B15 (recording fallback), B16 (request-id), B17 (Sentry releases).
- Plus DPDP floor: D1 (opt-out keywords), D2 (propagation), D3 (consent proof), D6 (grievance officer), D9 (24h window).

Outcome at day 30: a dramatically tighter product. NPS-movable.

Days 31–60 — Workflow depth

Theme: *the workflows your power users do 50 times a day get 50% faster.*

- Inbox composer: saved replies (W3), scheduled send (W4), reply-to-message (W5), AI draft tone controls (W7), idempotency (already shipped in 30d).
- Conversation list: snooze (W14), saved views (W15), SLA timer (W16), search (W17), agent presence (W18).
- Kanban: bulk actions (K10), saved filters (K11), inquiry-aging widget (K13), round-robin (K14).
- Call detail: real waveform (C2), inline coaching comments (C7), bookmarks (C8), playback speed (C10), keyboard shortcuts (C11).
- Calls list: pagination (already done) + smart views (C19, C22), trend column (C20).
- Templates: linter (T4), live preview (T5), CSV import for subscribers (T7).
- Backend: B3 (Celery retry), B5 (delivery state), B8 (rate limiting), B10 (per-tenant timezone), B12 (test mocks).
- Settings: S1–S3 (integration error states), S7 (Meta quality), S9 (webhook log).
- Compliance: D4 (bilingual notice), D5 (retention purge), D7 (audit log resource), D10 (template linter), D11 (delete grace).

Outcome at day 60: power-user workflows are buttery; integrations are observable; compliance posture is defensible.

Days 61–90 — Discoverability + scale

Theme: *the first-week experience and the long-tail polish.*

- Onboarding: O1 (skip steps), O2 (sample data), O5 (in-product checklist), O6 (test connection), O8 (granular consent), O9 (sample convo), O10 (WA team invite).
- Cross-cutting: X6 (command palette), X7 (empty states audit), X8 (global status bar), X10 (in-app help).
- Analytics: A1 (drill-down), A2 (compare period), A3 (per-property), A4 (drill-into theme), A7 (daily owner digest).
- Guest detail: G2 (revenue summary), G5 (quick actions), G7 (merge guests), G8 (delete grace).
- Mobile: M1–M7 (responsive audit).
- Tech debt: B9 (message partition strategy planning), B13/B14 (E2E + Celery tests), B18 (slow queries), B19 (Redis policy).
- Compliance: D12 (breach template), D13 (PII redaction in logs).

Outcome at day 90: a polished, discoverable product with infrastructure ready for the next 2× of customers.

Part V — How to measure the polish is working

Don't ship polish blindly. Instrument these so you know if it's landing:

Metric	Why it matters	Target by day 90
Median time-to-first-response (across channels)	Top operator pain	< 12 min
% of conversations with internal notes / @mentions used	Adoption of polish features	> 30% of active accounts
Saved-reply usage per active user / week	Power-feature adoption	> 8 per week
Inbox keyboard-shortcut events / user / week	Power-user signal	> 20 per week
Audit log undo-toast click rate	Validates the undo bet	4–8% of destructive actions
Calls auto-played from transcript click	Transcript-first works	> 60% of call detail sessions
Onboarding completion rate	Onboarding cost	> 50% (today: probably

(full 9 steps)		<30%)
Sample-data toggle → real-message switch	Activation	> 70% within 7 days
Median webhook delivery success rate	Reliability	> 99.5%
Tenant-NPS (rolling 30-day)	The thing that pays for polish	+15 vs today
Customer support tickets per active account / month	Reduced friction = fewer tickets	-30% vs today
Logo retention 90-day	The truth	+5 percentage points

Publish these to the team weekly. Polish without instrumentation is decoration.

Part VI — One uncomfortable principle to close

The biggest risk in a "polish phase" is that it feels less ambitious than "V2 features," and ambition-hungry teams will quietly start adding things. Resist this for a quarter. Polish that lands compounds: the same operator who would have churned at month 2 over a missing undo will, three months later, be your case study.

You already have a more capable product than you give yourself credit for. The job for the next 90 days is to make every existing feature feel inevitable — fast, reliable, observable, undoable. After that, ship V2 from a position of strength. Not before.

End.